



CSE461

Introduction to Robotics Lab

Lab No. : 06

Group : 01

Section : 06

Semester : Summer_2025

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Submitted Date: 20-08-2025

Submitted to - CSDM

1. Objectives:

The main objective of this lab is to create a complete system for controlling a DC motor's direction in real time using an analog joystick. This involves learning to interface the joystick with an Arduino to read its analog signals, properly implementing an L298N motor driver circuit to handle the motor's power requirements, and developing the necessary Arduino code to translate the joystick's input into precise motor commands. A key goal is to gain hands-on experience in diagnosing and troubleshooting the hardware and software to ensure seamless integration between the analog input and the motor output.

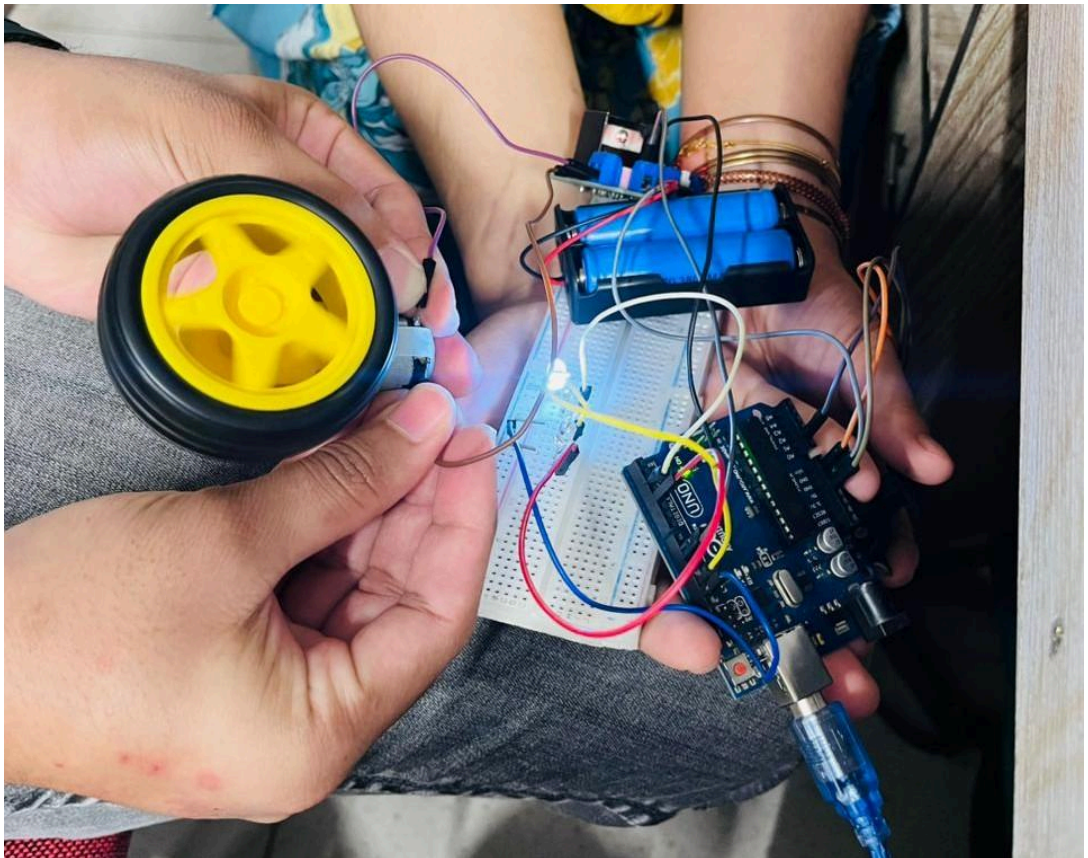
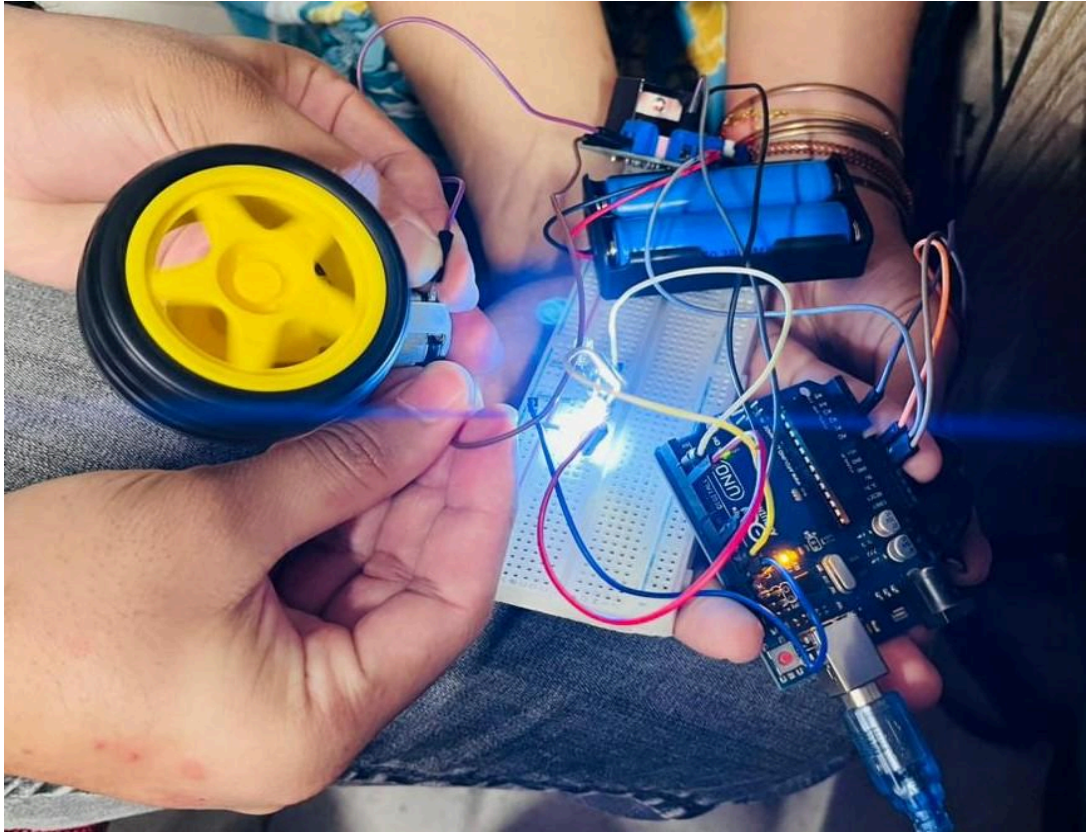
2. Equipments:

- Arduino Uno R3
- L298N Motor Driver Module
- 5-Pin Analog Joystick Module
- DC Motor (6V-9V)
- External Power Supply (Battery Pack)
- 3x LEDs (for the final task)
- Breadboard
- Jumper Wires

3. Experimental Setup:

(Picture + Explanation)

The experimental setup consists of a complete circuit where an Arduino serves as the brain, interpreting user commands from a 5-pin analog joystick connected to its analog pins (A0, A1). Because the Arduino cannot power the motor directly due to high current demands, it sends low-power directional signals from its digital pins (9, 10) to an L298N motor driver. This driver acts as a crucial bridge, taking power from a separate external battery pack and delivering it to the DC motor, causing it to spin forward or backward based on the joystick's movement. A critical part of the setup is the common ground connection, which links the Arduino and the motor driver's power supply to ensure the control signals work reliably.



4. Code: (If Applicable)

```
#define VRy A1
#define threshold 200
#define IN1 10
#define IN2 9

#define LED_FORWARD 2
#define LED_BACKWARD1 13
#define LED_BACKWARD2 4

void setup() {
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(LED_FORWARD, OUTPUT);
  pinMode(LED_BACKWARD1, OUTPUT);
  pinMode(LED_BACKWARD2, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  int y = analogRead(VRy);
  Serial.print("Y-axis: ");
  Serial.println(y);

  if (y < 512 - threshold) { // Forward
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);

    digitalWrite(LED_FORWARD, HIGH);
    digitalWrite(LED_BACKWARD1, LOW);
    digitalWrite(LED_BACKWARD2, LOW);

    Serial.println("Motor: Forward");
  }
  else if (y > 512 + threshold) { // Backward
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);
```

```

digitalWrite(LED_FORWARD, LOW);
digitalWrite(LED_BACKWARD1, HIGH);
digitalWrite(LED_BACKWARD2, HIGH);

Serial.println("Motor: Backward");
}
else { // Stop
digitalWrite(IN1, LOW);
digitalWrite(IN2, LOW);

digitalWrite(LED_FORWARD, LOW);
digitalWrite(LED_BACKWARD1, LOW);
digitalWrite(LED_BACKWARD2, LOW);

Serial.println("Motor: Stop");
}

delay(300);
}

```

5. Results (Output of the experiment):

The experiment resulted in a fully functional and responsive motor control system where the DC motor's direction was directly controlled by the joystick's Y-axis. The system provided clear visual feedback: pushing the joystick forward made the motor spin forward and lit up a single "forward" LED, while pulling it backward reversed the motor's rotation and turned on two separate "left and right" LEDs. When the joystick was released to its neutral position, the motor immediately stopped, and all LEDs turned off, confirming the 'otherwise' state. This behavior demonstrated the successful integration of the analog input, digital logic, and motor actuation as outlined by the lab task.

6. Discussions/Answers:

(May contain conclusions or learnings from the result along with problems faced and solving methodology.)

This lab was all about learning how a joystick can control a motor in real-time. The main takeaway was seeing how the signal travels: the joystick's movement is read by the Arduino as a number between 0 and 1023, and the Arduino then tells a motor driver what to do. A key lesson is that an Arduino can't power a motor on its own because it's not strong enough; it needs the L298N motor driver to act as a bridge to a more powerful source. This driver chip is also what allows the motor to spin both forward and backward. The process wasn't smooth, though, as some initial hassles were caused by a faulty Arduino and motor, which had to be replaced before the experiment would work. We

faced another issue, the digital pin3 on the Arduino wasn't working. Because of this, when the joystick was moved backward, two lights were supposed to turn on, but only one did. Later, we changed the connection to pin 13 and then a light on the board was turned on.